OSITO DRONE STEM ACADEMY

SERIES 11: AERODYNAMICS & FOUR FORCES OF FLIGHT - AGES 6-11 (GRADES 1-5)

THIS STEM COLORING BOOK BELONGS TO

YOUNG PILOT'S NAME / NOMBRE DEL PILOTO

MY DRONE'S CALLSIGN / NOMBRE DE MI DRON

CO-PILOT / GROWN-UP CO-PILOT NAME

Title: Vol. 501: Elena's Lift Force Balance at Extreme Weather Hurricane Tracking Center

Featured STEM Pilot: Elena (Junior Aerodynamicist & Wind Tunnel Pilot)

Flight Environment: Extreme Weather Hurricane Tracking Center - Target Age: Ages 6-11 (Grades 1-5)

Core STEM & Math Focus: Bernoulli Principle & Airfoil Pressure Differential

Includes: 8 Single-Sided Coloring Plates, 4 Vector STEM Activities, Flight Log & Certificate!

Explore Youth Drone STEM Camps, 3D Printing & Flight Simulators at ositodrone.com

Published by Osito Drone LLC - All Rights Reserved

OFFICIAL YOUTH DRONE SAFETY RULES & FAA TRUST NOTICE

SAFETY RULES FOR RECREATIONAL DRONE PILOTS

1

1. ALWAYS FLY WITH A GROWN-UP

Make sure a parent, teacher, or coach supervises every flight.

2

2. FLY IN OPEN, SAFE SPACES

Stay clear of people, pets, roads, trees, and active airport airspace.

3

3. KEEP DIRECT LINE-OF-SIGHT

Always keep your quadcopter where you can see it with your own eyes.

4

4. USE DUCTED PROPELLER GUARDS

Inspect all 4 ducted prop guards and battery latch before takeoff.

5

5. COUNT & CHECK PRE-FLIGHT MATH

4 propellers + 1 charged battery + 1 adult co-pilot = Safe Takeoff!

6

6. TAKE THE FREE FAA TRUST TEST

Learn official recreational sky rules at faa.gov/uas/trust with family.

OSITO DRONE LLC – PUBLISHING & LEGAL ATTRIBUTION

Osito Drone STEM Academy Curriculum Series - KDP Paperback Edition

Copyright (C) 2026 Osito Drone LLC. All rights reserved.

Published by Osito Drone LLC

Printed in the United States of America - Amazon KDP Print-on-Demand

Standard Edition: 8.5 x 11.0 in, 55# Uncoated White Interior Paper

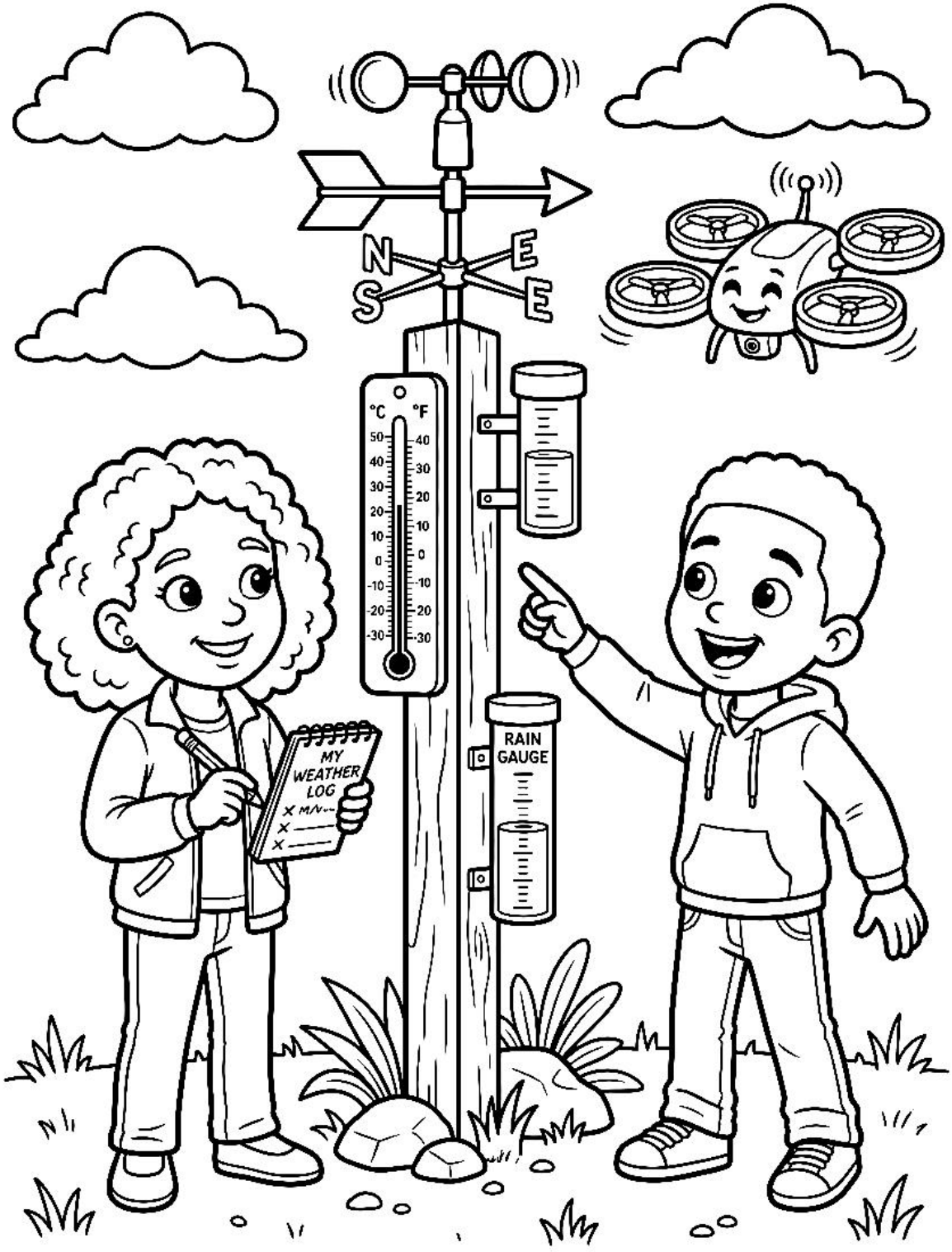
Publisher Contact: legal@ositodrone.com - Website: ositodrone.com

All characters, curriculum activities, and airframe schematics are property of Osito Drone LLC.

Designed to comply with FAA TRUST youth aviation educational benchmarks.

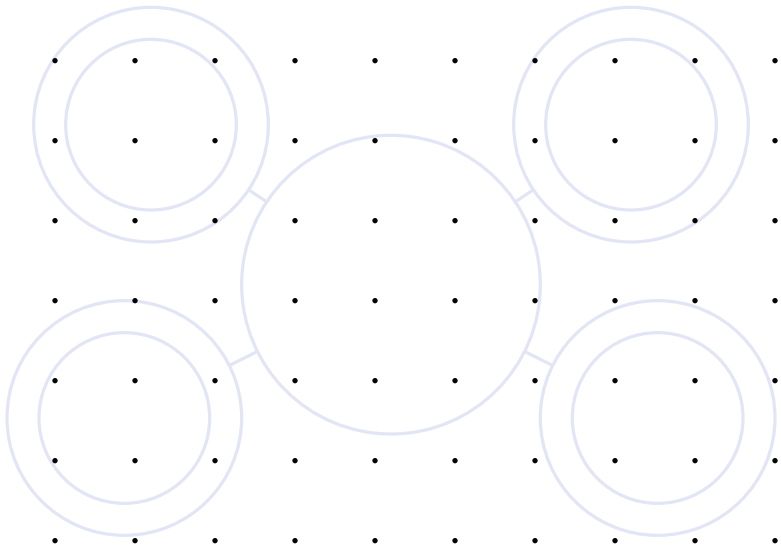
No part of this publication may be reproduced without prior written permission of Osito Drone LLC,

except for non-commercial educational classroom instruction by licensed educators.



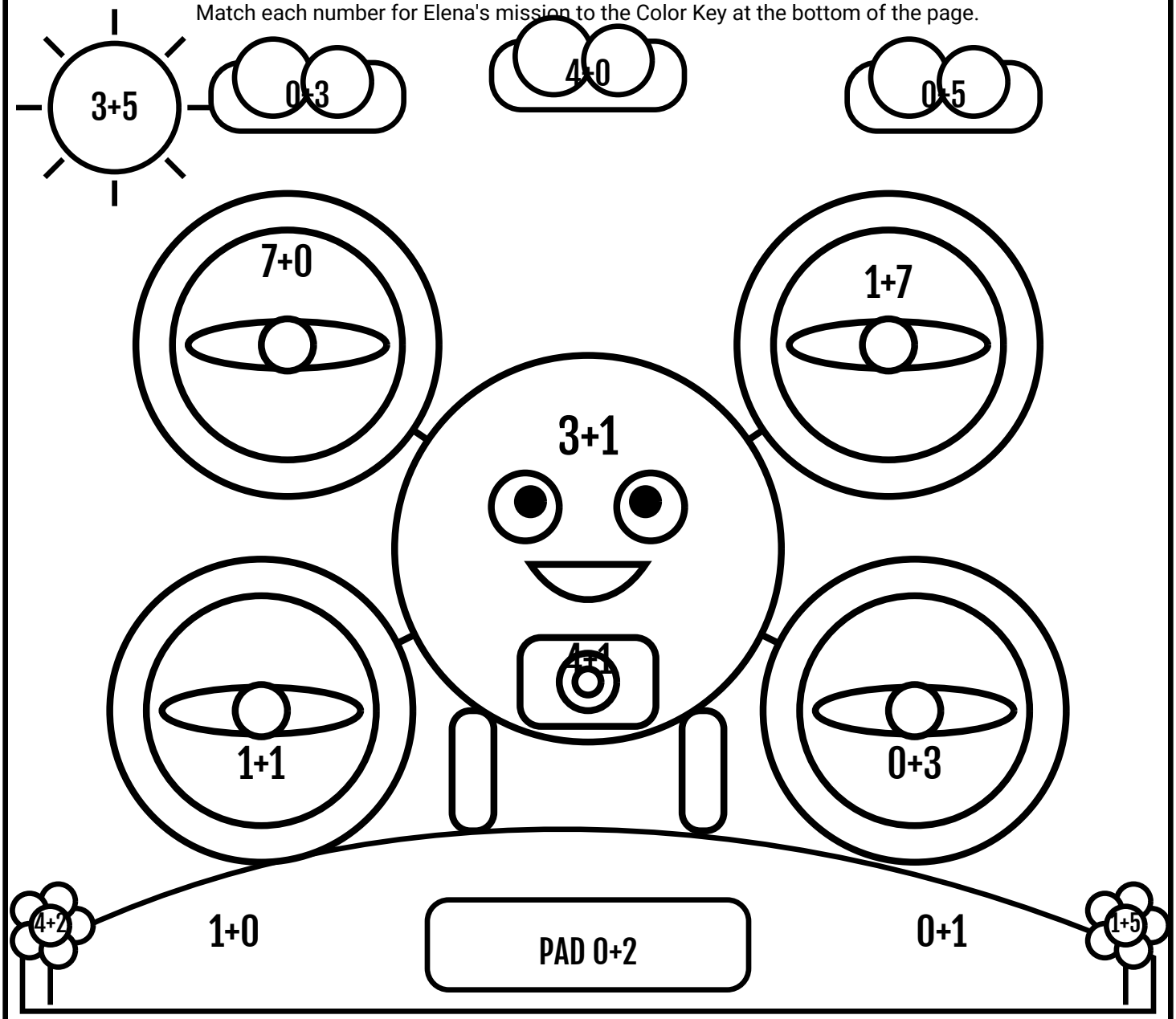
ELENA & PET DRONE – LIFT FORCE BALANCE

BLEED-PREVENTION BACKING PAGE 4 - BOOK #501 - FLIGHT LOG & SKETCH PAD



COLOR BY NUMBER: ELENA'S LIFT FORCE BALANCE

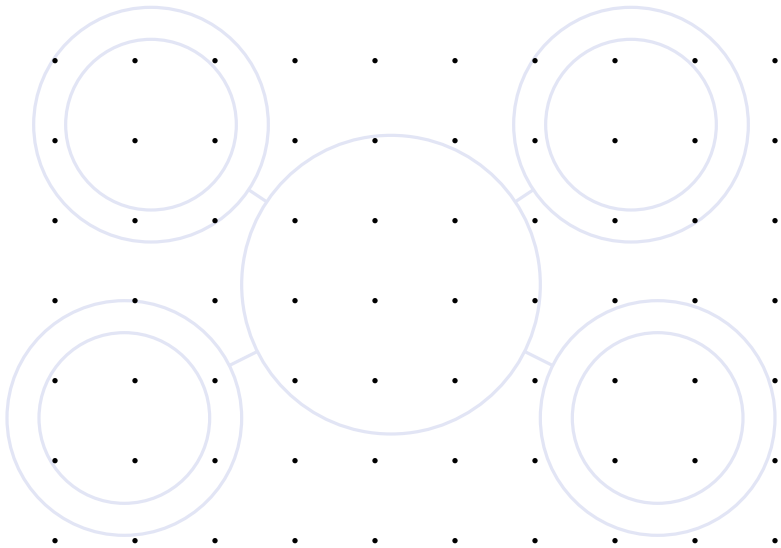
Match each number for Elena's mission to the Color Key at the bottom of the page.



NUMBER-BY-NUMBER COLOR KEY

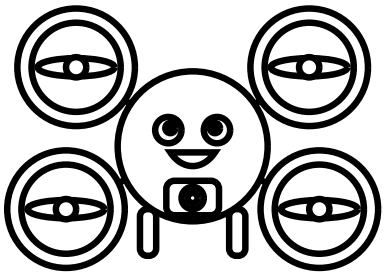
1 = RED	2 = BLUE	3 = YELLOW	4 = GREEN
5 = ORANGE	6 = PURPLE	7 = TEAL	8 = BROWN

BLEED-PREVENTION BACKING PAGE 6 - BOOK #501 - FLIGHT LOG & SKETCH PAD



DRONE MATH: ELENA'S LIFT FORCE BALANCE

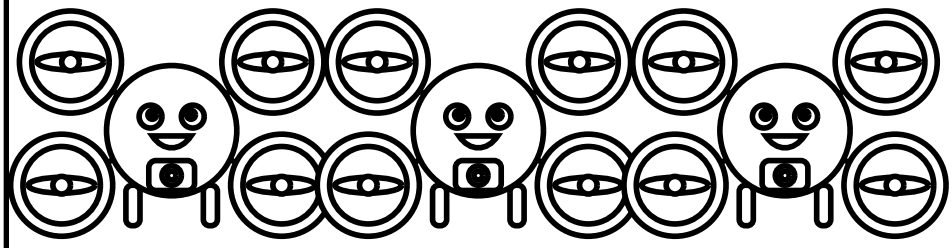
Every quadcopter has 4 ducted propellers! Solve Elena's flight problems and trace the numbers.



MISSION #1: ADD ROTORS

4 Props + 4 = 8

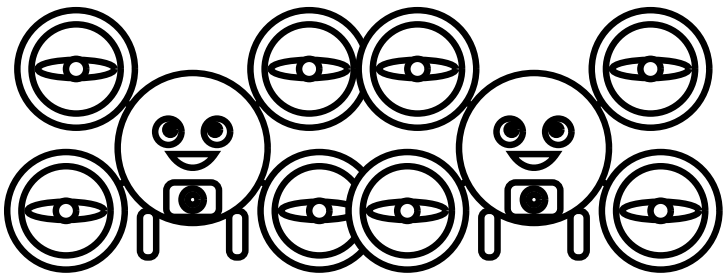
Color the drones & trace numbers!



MISSION #2: ADD ROTORS

12 Props + 5 = 17

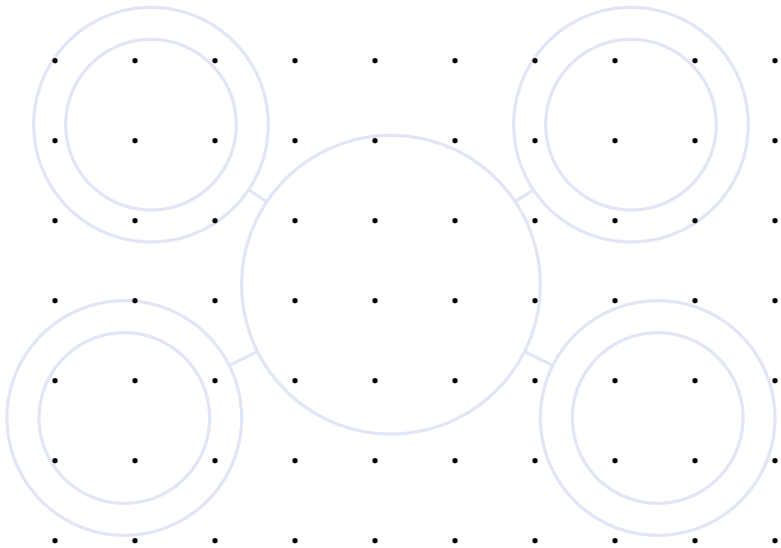
Color the drones & trace numbers!

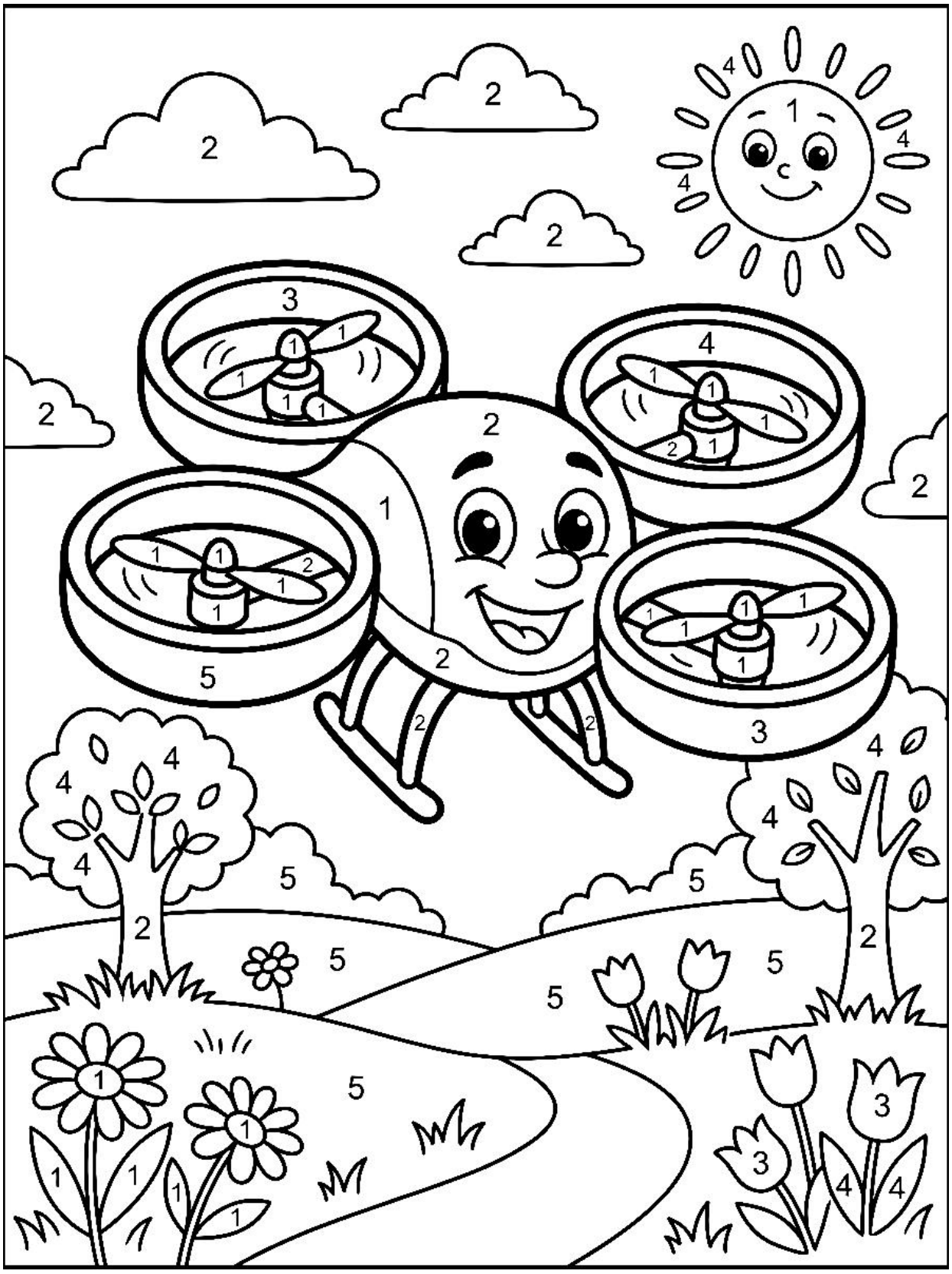


MISSION #3: ADD ROTORS

8 Props + 6 = 14

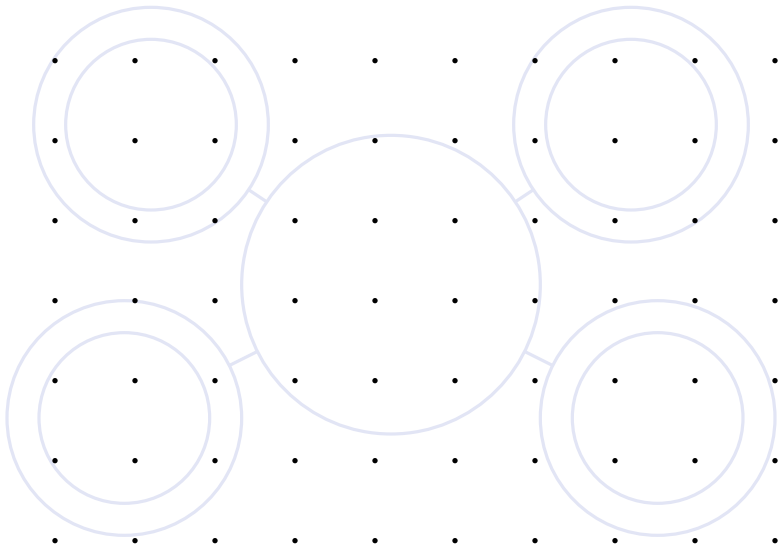
Color the drones & trace numbers!





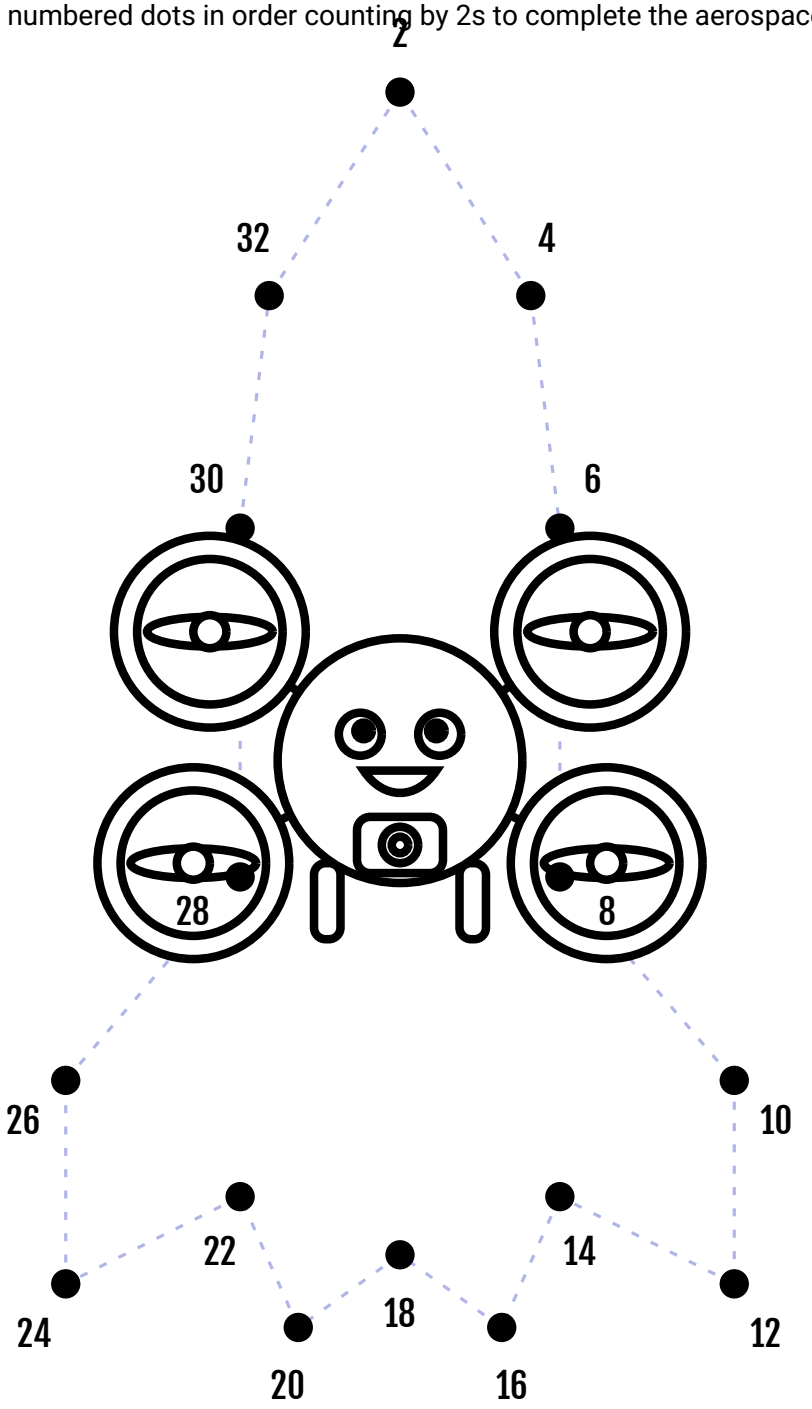
COLOR BY NUMBER & COUNT ROTORS WITH ELENA!

BLEED-PREVENTION BACKING PAGE 10 - BOOK #501 - FLIGHT LOG & SKETCH PAD



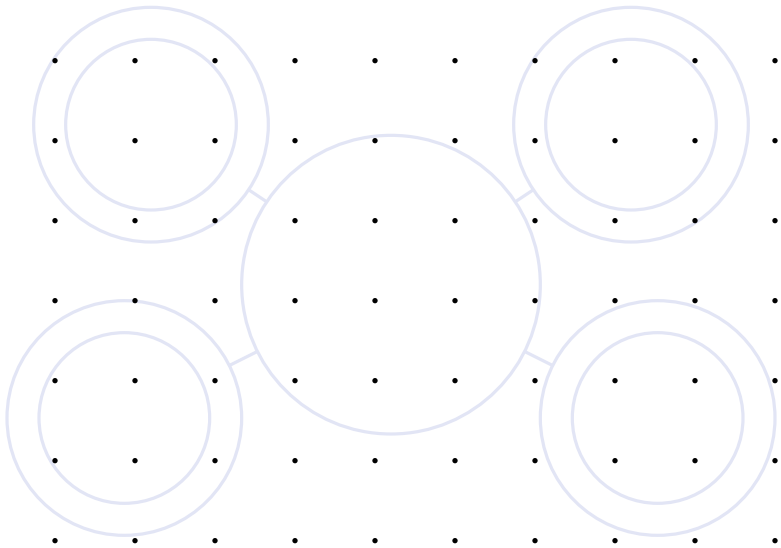
CONNECT THE DOTS: ELENA'S LIFT FORCE BALANCE

Connect the numbered dots in order counting by 2s to complete the aerospace blueprint!



BONUS FOR ELENA: You connected 16 dots counting by 2s! Final number = 32!

BLEED-PREVENTION BACKING PAGE 12 - BOOK #501 - FLIGHT LOG & SKETCH PAD



CODE-BY-NUMBER: SOLVE & COLOR PIXEL DRONE

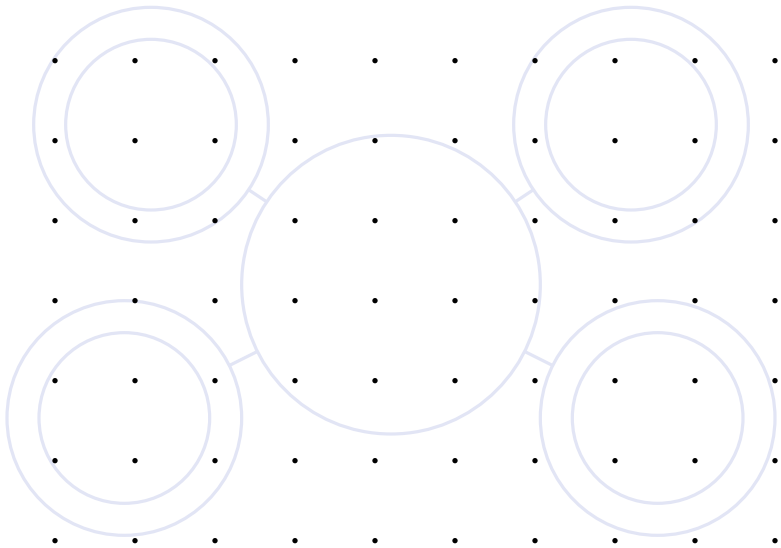
Solve or match the number in each grid square (A1-G7) using the Color Code Key below to reveal pixel art!

	1	2	3	4	5	6	7
A	4+1	2+3	4+1	1+0	5+0	5+0	2+3
B	5+0	2+3	1+0	2+1	0+1	4+1	4+1
C	1+4	1+4	1+1	0+2	0+2	0+5	0+5
D	4+1	0+5	2+0	1+2	2+0	1+4	0+5
E	2+3	0+1	1+1	2+0	2+0	0+1	3+2
F	0+1	0+1	3+0	0+1	1+2	0+1	0+1
G	2+2	1+3	3+1	4+0	1+3	0+4	2+2

PIXEL CODE COLOR KEY

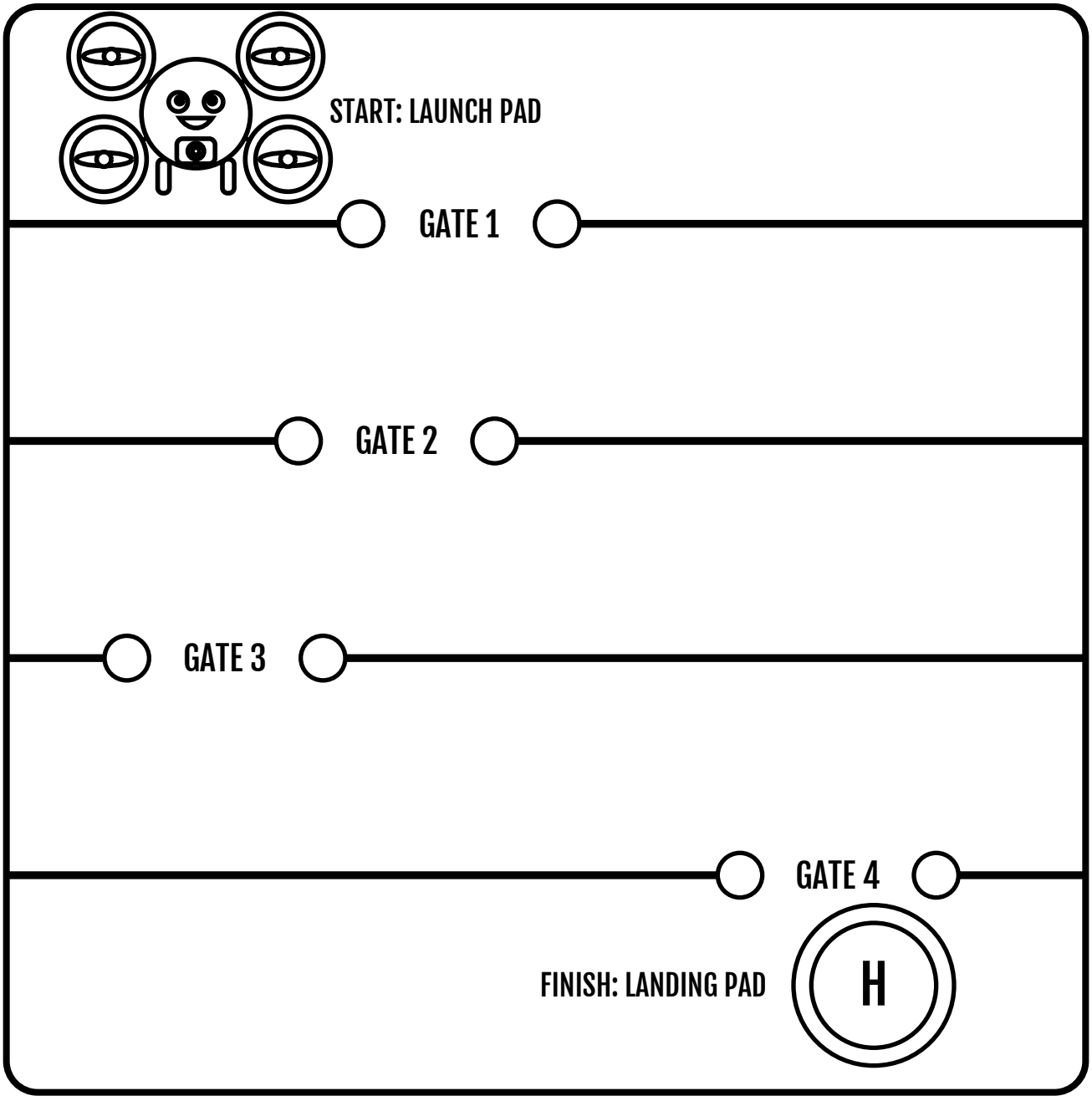
1 = RED 2 = BLUE / INDIGO 3 = GOLD / YELLOW 4 = GREEN 5 = LIGHT SKY BLUE

STEM Tip: Computers and drone cameras store pictures as grids of numbered pixels just like this!



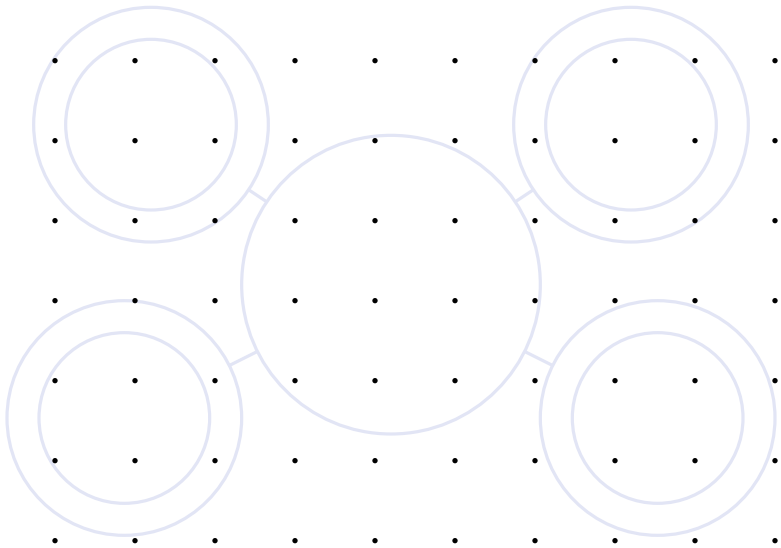
SAFE FLIGHT-PATH MAZE: GUIDE ELENA'S DRONE!

Draw a line from LAUNCH at top-left through open gates in order (1 -- 2 -- 3 -- 4) to LANDING PAD 'H'!



COLORING CHALLENGE: Color all 4 Gate Hoops different colors and draw arrows along your safe flight path!

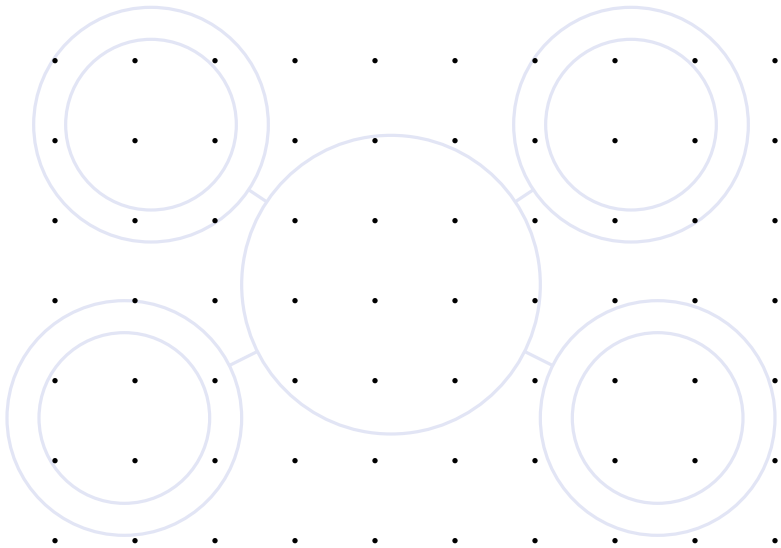
BLEED-PREVENTION BACKING PAGE 16 - BOOK #501 - FLIGHT LOG & SKETCH PAD





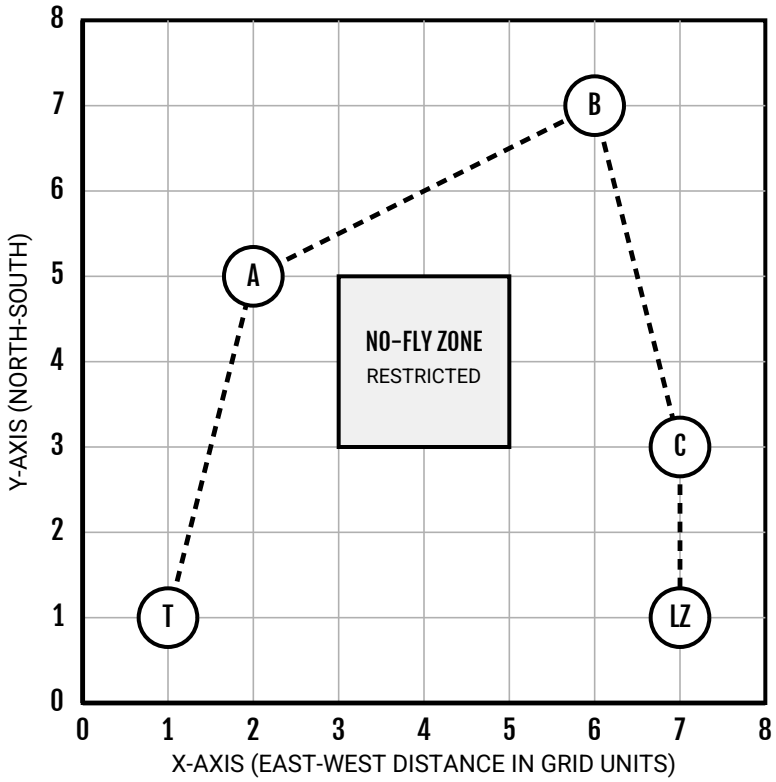
SAFE DRONE FLIGHT AT EXTREME WEATHER HURRICANE TRACKING CENTER

BLEED-PREVENTION BACKING PAGE 18 - BOOK #501 - FLIGHT LOG & SKETCH PAD



COORDINATE FLIGHT PLAN: ELENA'S WAYPOINT NAVIGATION

Plot waypoints on the (X, Y) grid for Elena's Lift Force Balance!



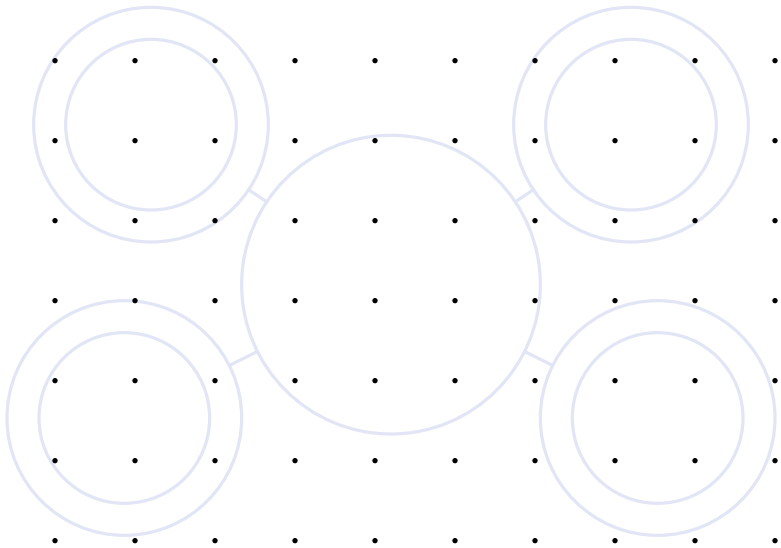
FLIGHT PLAN WAYPOINTS

WP T:	Takeoff Pad	(1 , 1)
WP A:	Elena Survey	(2 , 5)
WP B:	Waypoint Lift Force B	(6 , 7)
WP C:	Bridge Check	(7 , 3)
WP LZ:	Landing Zone	(7 , 1)

COORDINATE CHALLENGE:

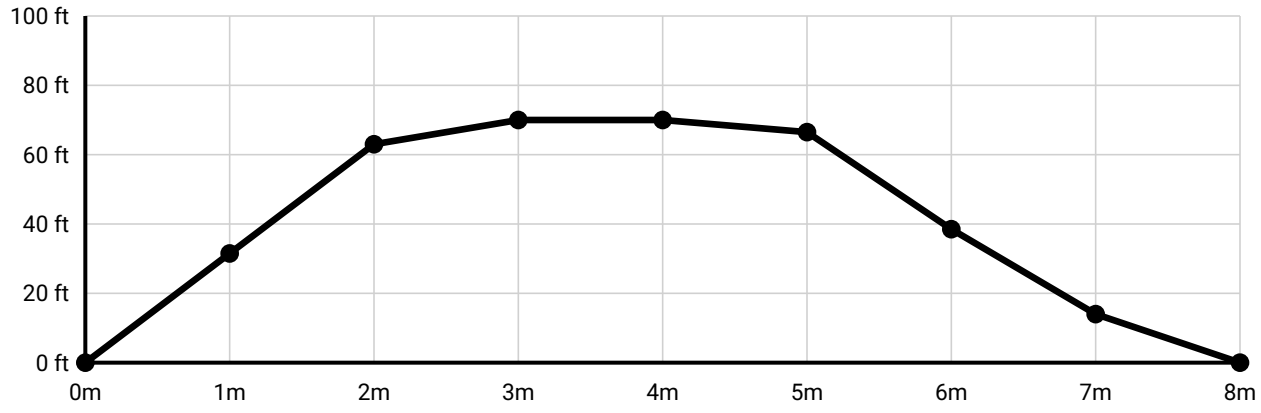
1. Waypoint B coordinates: (6 , 7)
2. Did flight avoid No-Fly Zone? [X] YES
3. Total flight legs flown: 4 legs

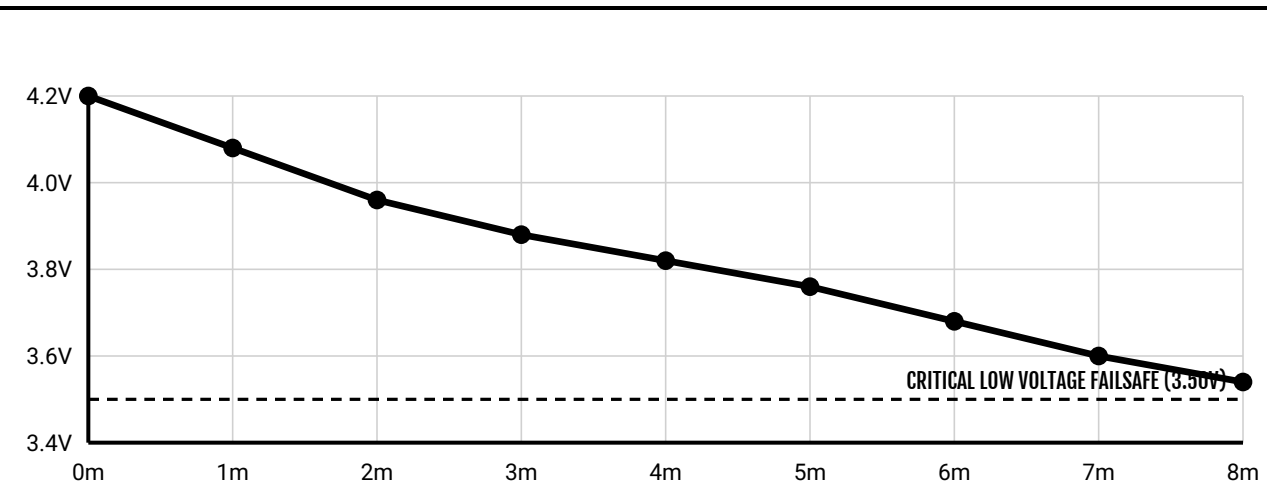
BLEED-PREVENTION BACKING PAGE 20 - BOOK #501 - FLIGHT LOG & SKETCH PAD



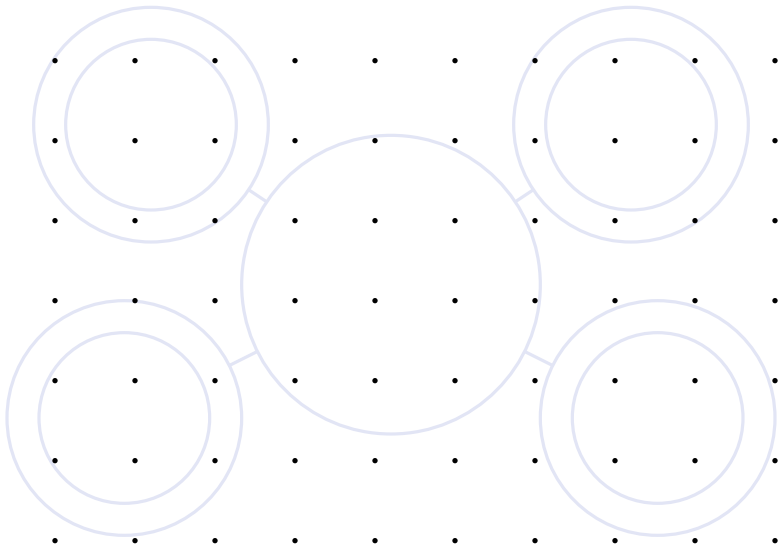
FLIGHT TELEMETRY: ALTITUDE & BATTERY DISCHARGE ENVELOPE

Analyze real flight telemetry data downloaded from the onboard BlackBox flight recorder.



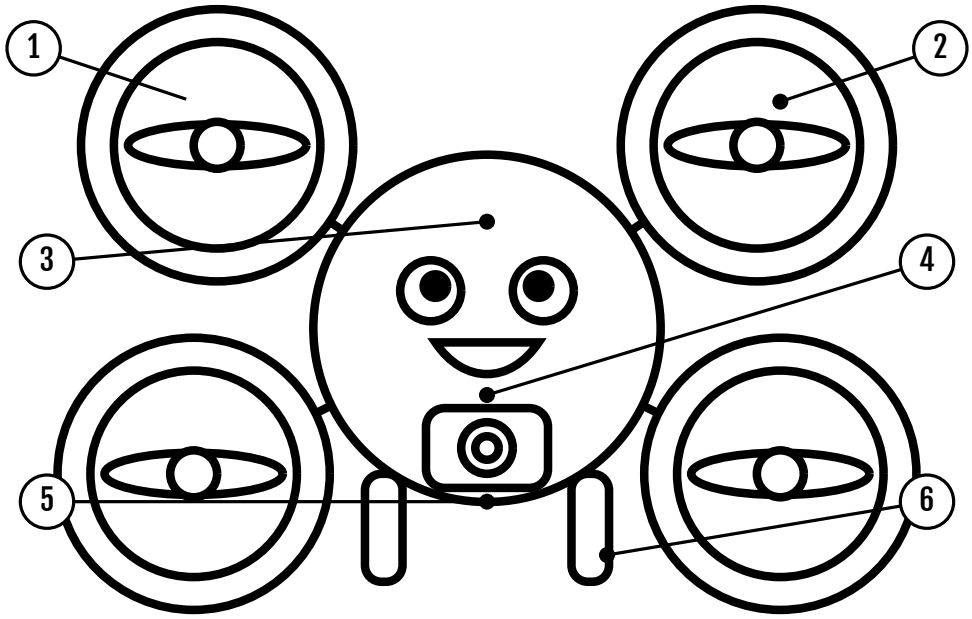


BLEED-PREVENTION BACKING PAGE 22 - BOOK #501 - FLIGHT LOG & SKETCH PAD



AEROSPACE BLUEPRINT: CUB-75 AVIONICS & AIRFRAME ANATOMY

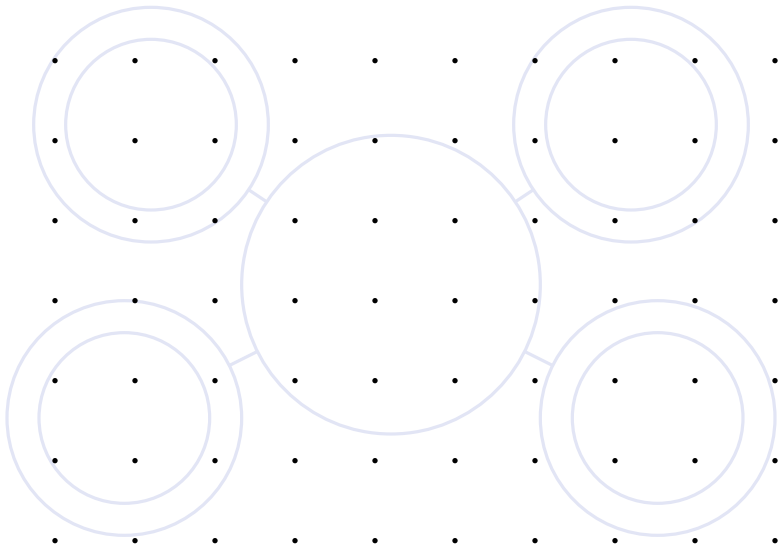
Study the mechanical airframe and avionics architecture of the Osito CUB-75 STEM training drone.



1	**Ducted Prop Guards**	Duct rings that shield propellers and provide 360-degree impact safety.
2	**Brushless DC Motors**	0802 19000KV motors that spin props at 30,000 RPM to produce vertical lift.
3	**Flight Controller (FC)**	Microprocessor & gyroscope that balances the drone 1,000 times every second.
4	**4-in-1 ESC Board**	Electronic Speed Controller regulating power sent from battery to each motor.
5	**Micro FPV Camera**	Ultra-wide lens sensor broadcasting real-time video to pilot FPV monitors.
6	**1S LiPo Battery**	3.8V High-Voltage Lithium Polymer rechargeable cell providing flight energy.

CUB-75 SPECS: Wheelbase: 75mm – AUW Weight: 39.5g – Motors: 0802 19000KV – Flight Time: 8.0 min

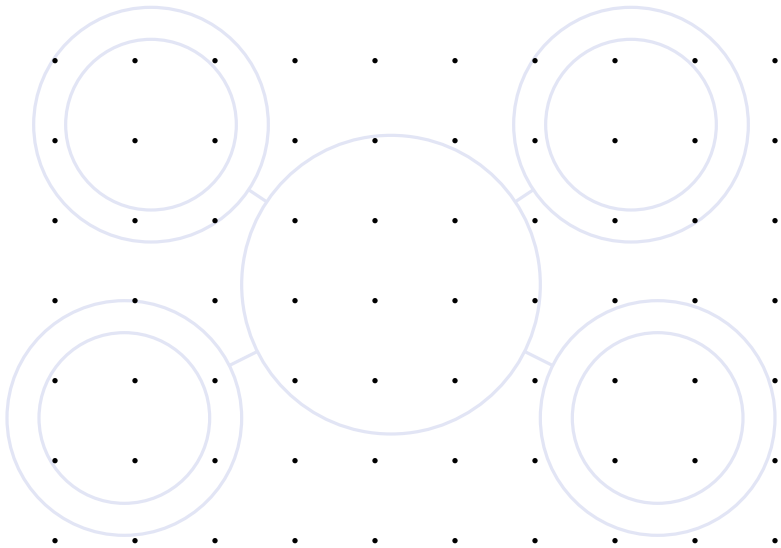
BLEED-PREVENTION BACKING PAGE 24 - BOOK #501 - FLIGHT LOG & SKETCH PAD



FLIGHT PHYSICS: THRUST-TO-WEIGHT & BATTERY ENDURANCE MATH

Apply real aeronautical engineering equations to calculate multirotor performance and endurance.

BLEED-PREVENTION BACKING PAGE 26 - BOOK #501 - FLIGHT LOG & SKETCH PAD



OFFICIAL OSITO JUNIOR PILOT FLIGHT MISSION LOG

RECORD YOUR ACTUAL FLIGHT MISSIONS AND COMPLETED STEM CHALLENGES

FLIGHT #	DATE	FLIGHT DURATION	MISSION OBJECTIVE	COACH SIGNATURE
Mission 1				
Mission 2				
Mission 3				

OSITO STEM PILOT CERTIFICATE OF ACHIEVEMENT

THIS CERTIFIES THAT JUNIOR PILOT

PILOT NAME HERE

HAS COMPLETED ALL STEM ACTIVITIES AND SAFETY PROTOCOLS IN

BOOK #501: LIFT FORCE BALANCE

Core Aeronautical Skill: Bernoulli Principle & Airfoil Pressure Differential

AND IS OFFICIALLY RECOGNIZED AS A CERTIFIED JUNIOR OSITO DRONE PILOT!

JUNIOR PILOT SIGNATURE

COACH / PARENT SIGNATURE & DATE

BLEED-PREVENTION BACKING PAGE 28 - BOOK #501 - FLIGHT LOG & SKETCH PAD

